

Enterprise ERP Prompt – Reel Inventory Management System (MERN)

Project Context:

You are building an Enterprise Resource Planning (ERP) module for a corrugation box manufacturing company, focused on Reel Inventory Management. This is a production-grade system with real-world inventory tracking logic.

User Role:

Single role – Inventory Operator. JWT-based authentication is mandatory.

Tech Stack:

Frontend: React (Vite), Tailwind CSS

Backend: Node.js, Express

Database: MongoDB

Authentication: JWT

Reel Identification:

Each reel type is uniquely identified using a composite key:

Size + GSM + Shade

Reel Properties:

- Size
- GSM
- Shade
- Auto-generated uniqueReelCode
- CreatedAt

Duplicate reel types are not allowed.

Inventory Tracking:

All quantities are measured in KG.

Inventory is transaction-based.

Daily Tracking:

- Opening Stock
- Daily Inward (purchase/production)
- Daily Usage
- Bit Reel (leftover) handling

Calculation Logic:

Opening Stock + Inward - Usage = Closing Stock

Negative stock is not allowed.

Database Collections:

Users

Reels

ReelTransactions

Dashboard:

Excel-like table view with search, sorting, and clickable rows.

Reel Detail View:

Transaction history with inward and usage entry.

Auto-calculation of remaining stock.

Add New Reel:

Sidebar-based form.

Validation and duplicate prevention required.

APIs:

Authentication

Create Reel

Fetch Reels

Reel Details

Inward Transaction

Usage Transaction

Non-functional Requirements:

Responsive UI

Clean architecture

Error handling

ERP-grade scalability

Final Instruction:

Build the complete Reel Inventory Management ERP module end-to-end following these specifications exactly.